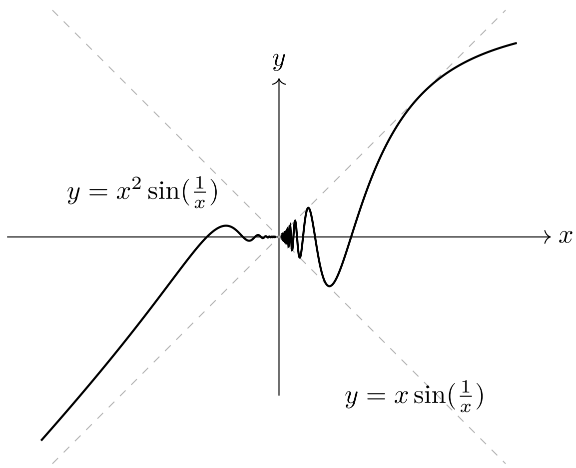
0.1 The function $x^a \sin(|x|^{-c})$

Definition 0.1.1. Define a map as: for $a \in \mathbb{R}$ and $c > 0$,

$$f_a^c : [-1, 1] \to \mathbb{R} : x \mapsto \begin{cases} 0 & \text{if } x = 0 \\ x^a \sin(|x|^{-c}) & \text{if } x \neq 0 \end{cases}$$



Equivalent Condition Table	f_a^c	f'
Continuous	$a > 0$	$a > 1 + c$
Bounded	$a \geq 0$	$a \geq 1 + c$
Differentiable at $x = 0$	$a > 1$	-
Bounded Variation	$a > c$	-

Lemma 0.1.2. The derivative of $f_a^c(x)$ is given by:

$$\frac{d}{dx} f_a^c(x) = \begin{cases} \lim\limits_{t \to 0} t^{a-1} \sin\left(\frac{1}{|t|^c}\right) & \text{if } x = 0 \text{ (provided the limit exists)} \\ a x^{a-1} \sin\left(\frac{1}{x^c}\right) - c x^{a-c-1} \cos\left(\frac{1}{x^c}\right) & \text{if } x > 0 \end{cases}$$

0.2 Typical Examples

Proposition 0.2.1. f_1^1 is continuous, **Not** differentiable at $x=0$, **Not** of bounded variation, and does **Not** have a bounded derivative.

Proof. Since $\left|x\sin\left(\frac{1}{x}\right)\right| \le |x| \to 0$ as $x \to 0$, f_1^1 is clearly continuous.
Meanwhile, $f'(0)$ exists if and only if $\lim_{t\to 0}\frac{f(t)-f(0)}{t} = \lim_{t\to 0}\sin\left(\frac{1}{t}\right)$ exists.
However, this limit does not exist, so f_1^1 is not differentiable at $x=0$. To show that f_1^1 is not of bounded variation, we construct a partition P_n as:

$$P_n \overset{\text{def}}{=} \left\{0, \frac{2}{n\pi}, \frac{2}{(n-1)\pi}, \dots, \frac{2}{2\pi}, \frac{2}{\pi}, 1\right\}$$

Then, the variation satisfies:

$$V(f_1^1, P_{2n+1}) \ge \sum_{k=0}^{n-1}\frac{2}{(2k+1)\pi} = \frac{2}{\pi}\sum_{k=0}^{n-1}\frac{1}{2k+1} \to \infty \quad \text{as } n \to \infty.$$

Therefore, f_1^1 is not of bounded variation. Similarly, f_n^n $(n \in \mathbb{N})$ are not of bounded variation. □

But, in this way, we cannot determine whether $f_2^1(x) = x^2\sin\left(\frac{1}{x}\right)$ is bounded variation or not.

Proposition 0.2.2. If $f:[a,b]\to\mathbb{R}$ is differentiable and the derivative f' is bounded, then f is of bounded variation.

Proof. For any partition P on $[0,1]$, by the Mean Value Theorem, there exists $t_i \in (x_{i-1}, x_i)$ such that

$$V(f,P) = \sum_{i=1}^{n}|f(x_i) - f(x_{i-1})| = \sum_{i=1}^{n}|f'(t_i)|\cdot\Delta x_i \le \sum_{i=1}^{n}\sup|f'|\cdot\Delta x_i = \sup|f'|\cdot 1$$

Since P was arbitrary, f is bounded variation. □

Corollary 0.2.3. f_2^1 is of bounded variation on $[-1,1]$.

Proposition 0.2.4. f_a^c is bounded variation on $[0,1]$ if and only if $a > c$

Proof. Suppose that $a \le c$. Then, simlilarly in the case of f_1^1, take a partition P_n on $[0,1]$ as:

$$P_n \overset{\text{def}}{=} \left\{0, \sqrt[c]{\frac{2}{n\pi}}, \sqrt[c]{\frac{2}{(n-1)\pi}}, \dots, \sqrt[c]{\frac{2}{2\pi}}, \sqrt[c]{\frac{1}{n\pi}}, 1\right\}$$

Then,

$$V(f_a^c, P_{2n+1}) \ge \sum_{k=0}^{n-1}\left(\frac{2}{(2k+1)\pi}\right)^{\dfrac{a}{c}} \to \infty \quad \text{as } n\to\infty$$

because $a \le c$ implies $\dfrac{a}{c} \le 1$. (In case of $a > c$ will be continue) □